

Unmasking Readmission

Cost-Sensitive, Transparent Knowledge Discovery for 30-Day
Hospital Readmission in Diabetic Patients

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CP8305 - Knowledge Discovery

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- **Industry Challenge:** Inability to predict short-term patient readmissions leads to:
 - Misdiagnosed severity
 - Suboptimal resource allocation
 - Financial penalties under HRRP¹
- **Project Goal:** Predictive and Perscriptive analysis of 30-day readmission risk in diabetic patients
 - Transparent methods, for trust
- **Expected Impact:**
 - Methods which are both interpretable and performative
 - Actionable insights for decision-making

¹Strack et al. 2014.

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- **Source:** Diabetes 130-US Hospitals dataset (1999-2008), Cerner Health Facts.
- **Target Variable Formulation:** Binary classification aligning with HRRP penalty windows:
 - **Positive (High-Risk):** < 30 days readmission (Minority class: 11%)
 - **Negative (Low-Risk):** > 30 days or NO readmission

- **Identified Gaps to Address:**

- “Black-box” ensemble models (e.g., XGBoost) lack the clinical transparency required for actionable healthcare decisions.

- **Data Manipulation Hurdles:**

- **Class Imbalance:** Only 11% of records for the readmission case.
- **Sparsity:** Many missing, non-estimatable values
- **High Cardinality:** 100's of medical codes; domain knowledge to combine.
- **Leaky Variables:** Patient codes non-unique; some discharges indicate mortality.

- **Existing Research Lacks:**

- punishment of false negatives²
- inpatient vs. outpatient medications³
- focus on transparent models⁴
- production of rule sets for clinical use⁵

²Rubin 2018.

³Ibid.

⁴Liu et al. 2024.

⁵Ibid.

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- **Feature Engineering:** Proposing new clinical variables.
- **Addressing Imbalance:** Accurately model the minority class via:
 - SMOTE
 - Cost-Sensitive Reweighting

Modeling: Interpretability vs. Performance

- **Our Approach:** Benchmark top models and interpretable, rule-based algorithms.
- **Candidate Models/Systems:**
 - *Transparent Base:* Logistic Regression, Decision Trees.
 - *Rule-Induction:* PRISM / Association Rules (human-readable logic).
 - *Black-Box Top Performer:* XGBoost.
 - *SHAP:* for black-box model interpretation.

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Why Accuracy Isn't Enough

- **Metrics/Systems:**

- *Stratified CV*: avoid leakage.
- *AUPRC*: for rare events
- *Kappa*: for dataset skew
- *Brier Score*: for true clinical likelihood

The Cost-Sensitive Framework

- **False Negative (Missed High-Risk):** High Cost. Patient readmits within 30 days, triggering heavy HRRP financial penalties and resource strain.
- **False Positive (Unnecessary Intervention):** Low-to-Moderate Cost. Enrolling a patient in preventive care (e.g., follow-up calls) who wouldn't have readmitted.
- **Goal:** Optimize the threshold to minimize overall financial and operational cost, not just classification error.

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- **Actionable Drivers:** [Placeholder for core indicators from 04_analysis_findings, e.g., HbA1c changes, specific comorbidities].
- **Clinical Application:** Transitioning from “black-box scoring” to transparent risk triggers enables targeted preventive resource allocation at discharge.
- **Recommendation:** Deploy an interpretable rule-set overlay to flag high-risk discharges, balancing intervention costs against HRRP penalty risks.

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Conclusion & Next Steps

- **Summary:** Shifted focus from raw prediction to Cost-Sensitive Knowledge Discovery, providing interpretable rules to mitigate HRRP readmission penalties.
- **Next Steps in Analysis:** Complete integration of specific KD techniques and formalize the custom evaluation matrix. Include findings from Unsupervised constraints.

Thank You!

Questions?

Project Repository: github.com/chrisindris/CP8305FP

